

Atmospheric Super-Resolution with Neural Operators

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Introduction:

- Climate emulators with coarse resolution data succeed in capturing dynamics but suffer from the loss of information.
- In this work, machine learning based super resolution models are explored to bypass the computing resources barrier for high resolution climate emulators.
- Two methods were investigated: CSRO with a backbone of Continuous Vision Transformer (CViT) and SSRO with Spherical/Fourier Neural Operators (S/FNO).

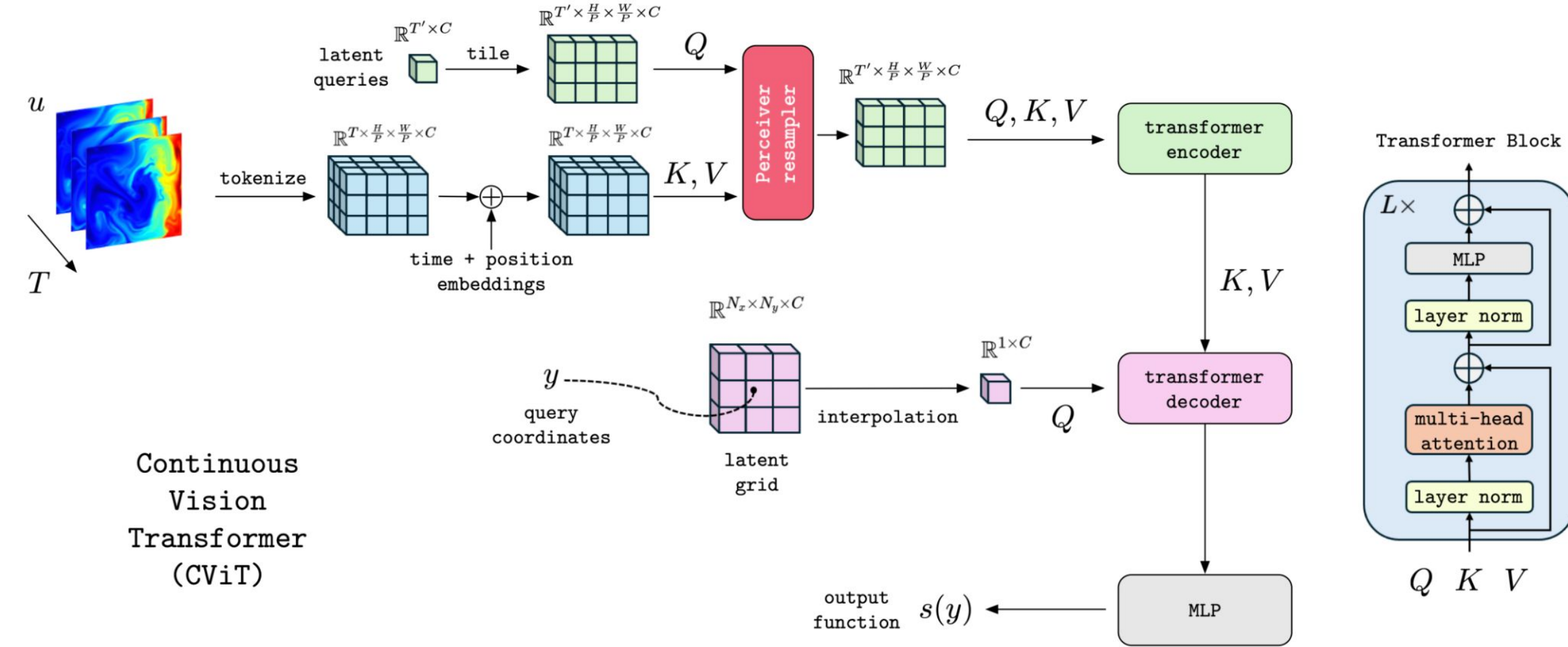


Fig 1: Schematic of CViT (Wang, et. al.), the backbone of Continuous Super Resolution Operator (CSRO)

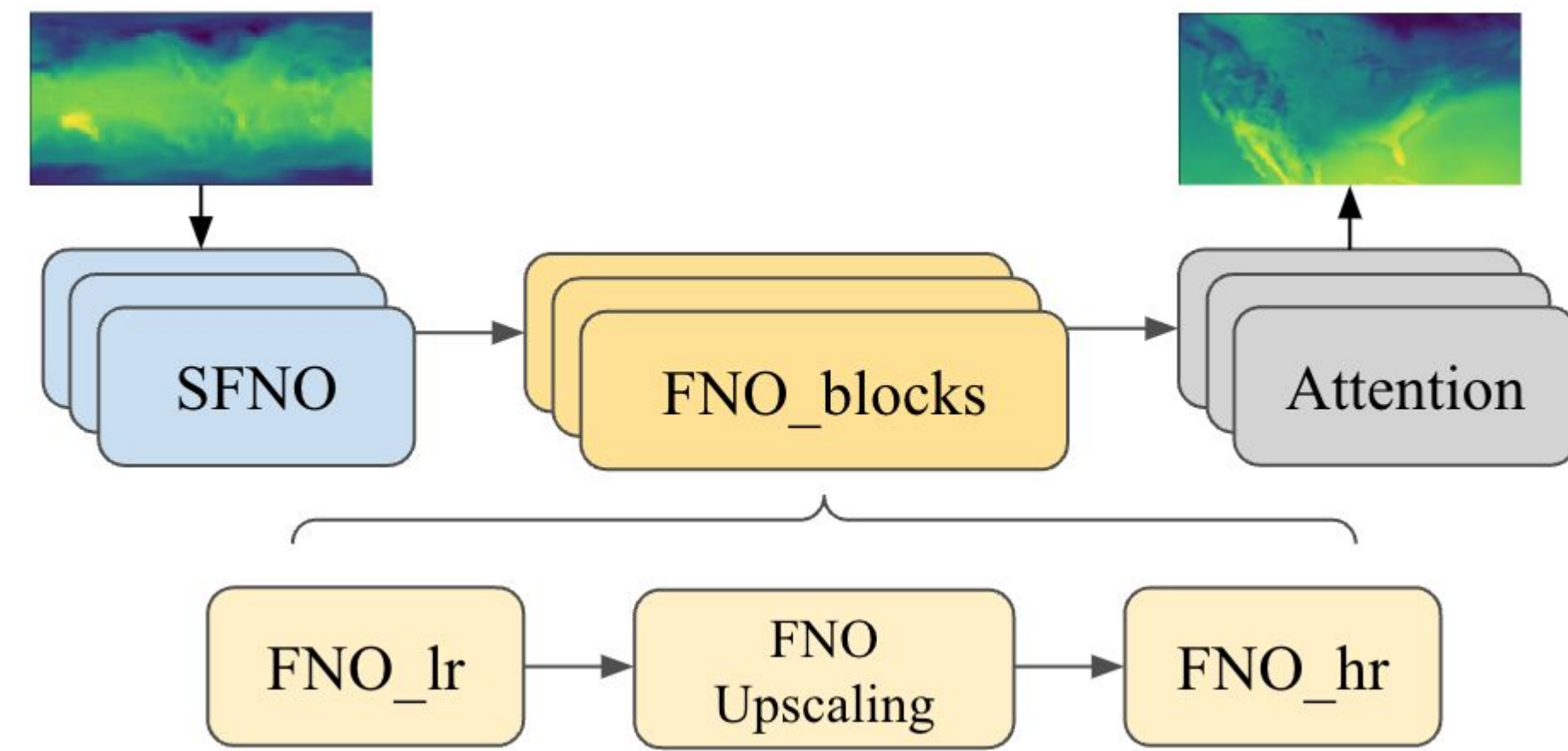


Fig 2: Schematic of Spectral Super Resolution Operator (SSRO) with SFNO and FNO backbones

Power Spectrum:

The power spectrum indicates the energy distribution among the atmospheric signal frequency component and shows the contribution of the large-scale flows and smaller turbulent patterns. Both models demonstrate promising performance on capturing the power spectrum of the ground truth ERA5.

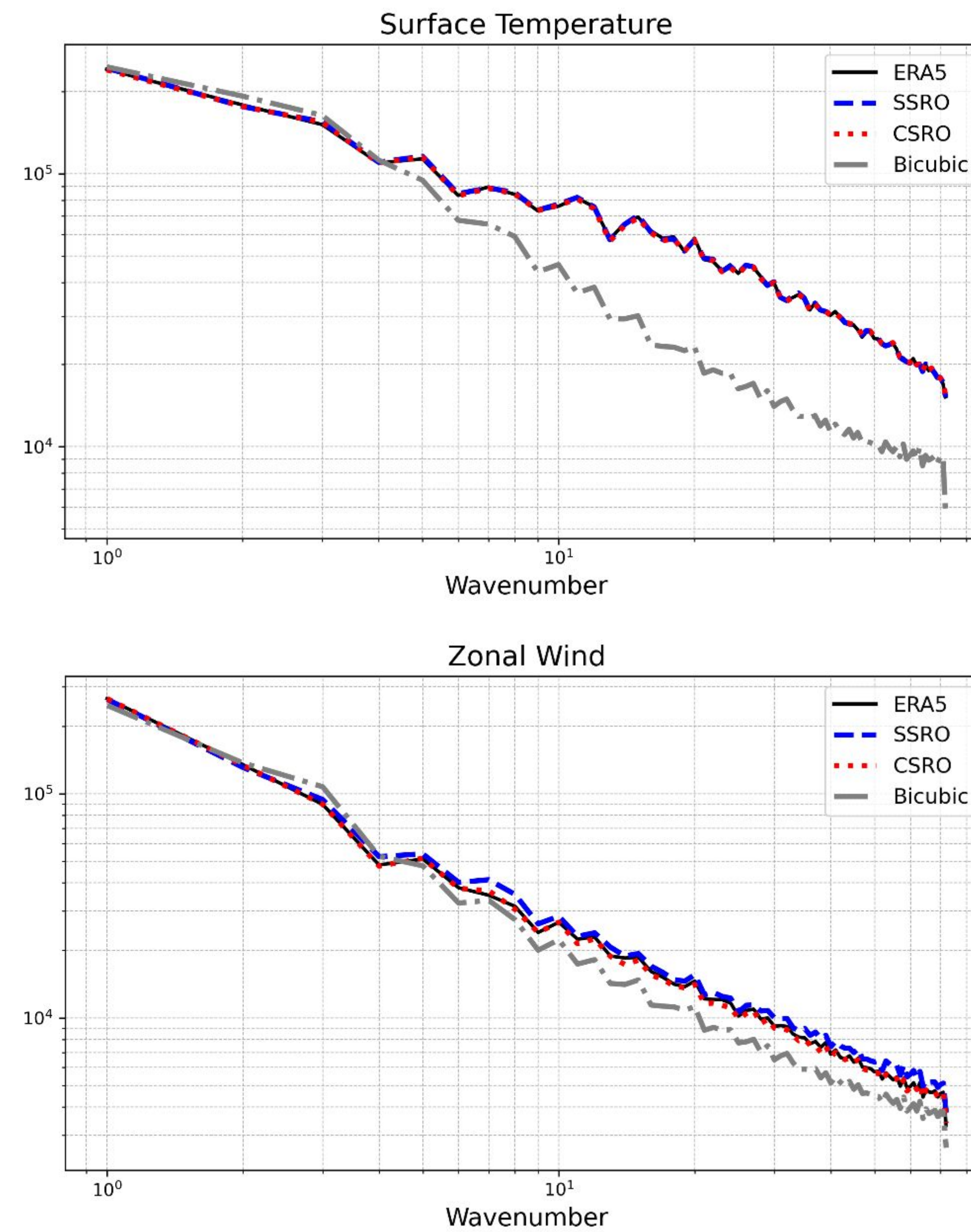


Fig 4: Power spectrum of SSRO, CSRO, ERA5, and Bicubic interpolation

Return Periods:

The ability of capturing and generating extreme events from the coarse resolution climate emulator LUCIE (Guan, et. al.) is one of the major interests for a super resolution model to recover. Along with other dynamics metrics, return period is an indicator for the ability of the super resolution model to preserve the dynamics.

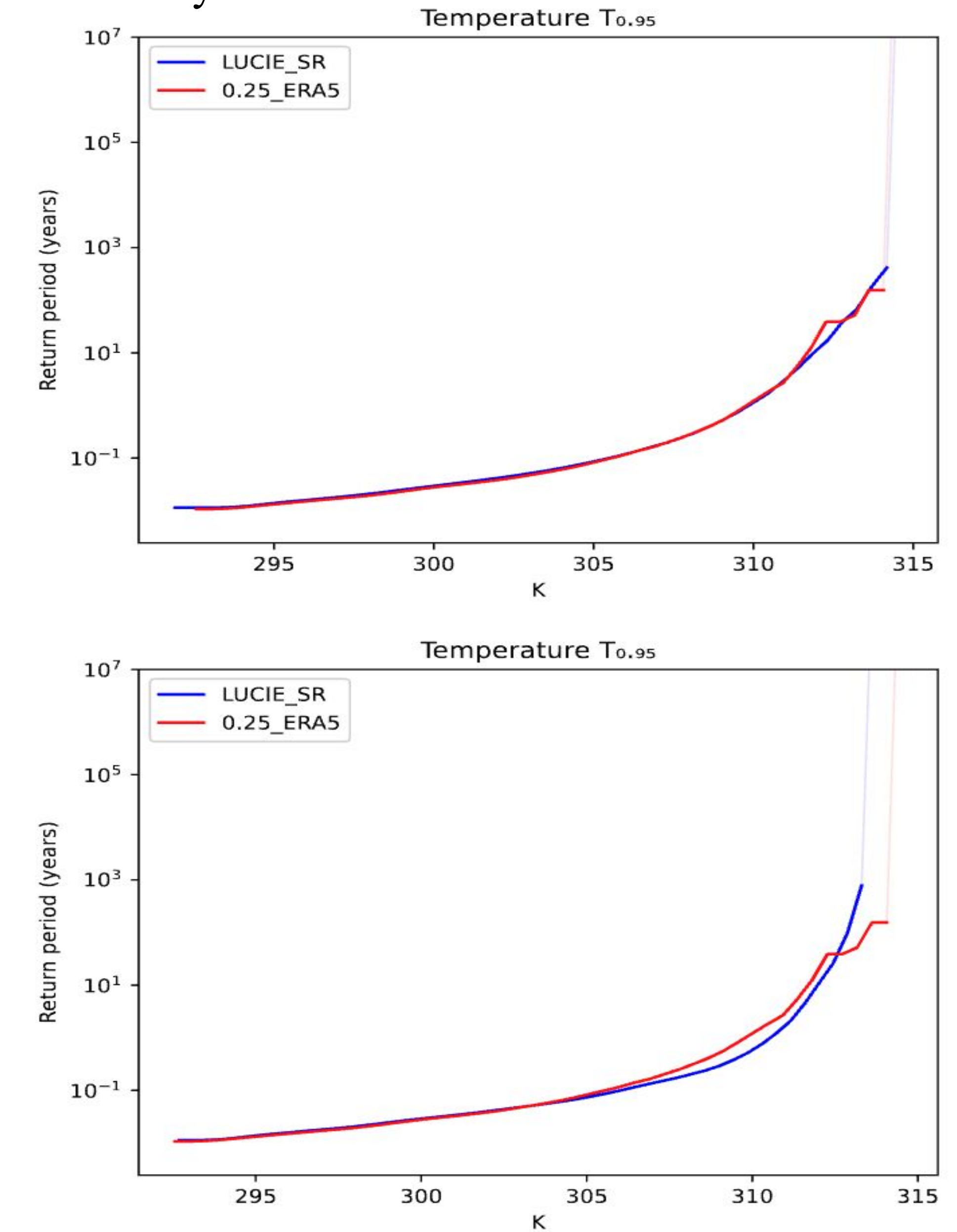


Fig 5: Return periods of SSRO AND CSRO super-resolved LUCIE inference with respect to ERA5

Results:

The models are trained with input of 3.75° ERA5 and output of 0.25° ERA5. Both models output CONUS areas. CSRO takes CONUS area as input and SSRO takes the full globe. The models are trained with variables: surface temperature, specific humidity at sigma level 7, zonal wind and meridional wind at sigma level 3, and precipitation. Temperature and zonal wind are shown here.

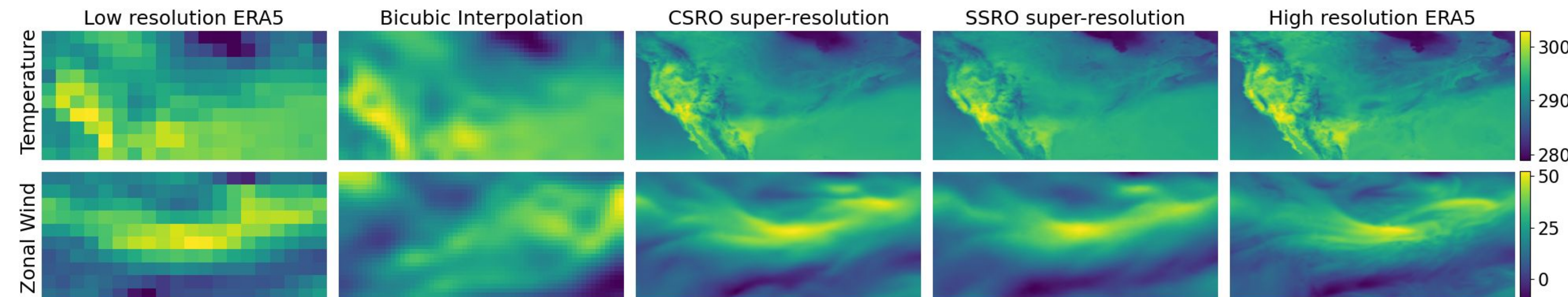


Fig 3: Super-resolution result of SSRO and CSRO

Discussion:

- Both methods investigated in this work have a similar performance and both approaches are under further development.
- Further research is needed to improve the deterministic models' ability to better capture the wind structure.
- We will conduct a more comprehensive evaluation with climatology metrics.

Reference:

Wang, S.; Seidman, J. H.; Sankaran, S.; Wang, H.; and Paris, G. J. P. 2024. Cvit: Continuous vision transformer for operator learning. *arXiv preprint arXiv:2405.13998*, 3.

Guan, H.; Arcomano, T.; Chattopadhyay, A.; and Maulik, R. 2024. LUCIE: A Lightweight Uncoupled Climate Emulator with long-term stability and physical consistency for $O(1000)$ -member ensembles. *arXiv:2405.16297*.